Bhavya Tulasi Tallapaneni
Data Engineer

Phone: 614-822-1429
Email: tallapanenib0@gmail.com

Professional Summary:

- Professional **Data Engineer** with **4+ years** of technical expertise in all phases of Software development cycle **(SDLC)** primarily in **Bigdata and Cloud computing areas.**
- Setting Up **AWS** and **Microsoft Azure with Databricks** Managing the **Machine Learning Lifecycle.**
- Experience in using **IDEs** like **Eclipse, IntelliJ IDE, PyCharm IDE, Notepad ++, and Visual Studio** for development.
- Experience on Migrating SQL database to **Azure data Lake, Azure data lake Analytics**, **Azure SQL Database, Data Bricks** and **Azure SQL Data warehouse** and controlling and granting database access and Migrating On premise databases to **Azure Data Lake store** using Azure Data factory.
- Experience in **Cloud Computing (AWS & Azure) and Big Data** tools like **Hadoop, HDFS, Map - Reduce, Hive, HBase, Spark, Spark Streaming, Azure Cloud, Amazon EC2, DynamoDB, Amazon S3,Kafka, Flume, Avro, Sqoop, Airflow PySpark.**
- Hands-on experience in developing and deploying enterprise-based applications using major Hadoop ecosystem components like **Spark MLlib, Spark GraphX, Spark SQL, Kafka.**
- Experience with Spark to improve efficiency of existing algorithms using Spark Context, Spark SQL, Spark MLlib, Data Frame, Pair RDD and Spark YARN.
- Experience in building **Data pipelines** for Real Time streaming data and Data Analytics using Azure cloud components like **Azure Data Factory, HDInsight (spark cluster), Azure ML Studio, Azure stream Analytics, Azure Blob Storage, Microsoft SQL DB, Neo4j (Graph DB).**
- Proficient with **Spark Core, Spark SQL, Spark MLlib, Spark GraphX, and Spark Streaming** for processing and transforming complex data using in-memory computing capabilities written in **Scala.**
- Experience in application of various data sources like **Oracle SE2, SQL Server, Flat Files** and Unstructured files into a **data warehouse.**
- Experience in using **Sqoop** to migrate data between **RDBMS, NoSQL databases and HDFS.**
- Experience working in **SQL Server** and **My SQL database.**
- Experience working in reading Continuous **json** data from different source systems using **Kafka** into **Databricks Delta** and processing the files using Apache Structured streaming, **PySpark** and creating the files in parquet format.
- Experience in working with different data formats **CSV, JSON, and Parquet.**
- Strong in **Data Warehousing** concepts, Star schema and Snowflake schema methodologies, understanding Business process/requirements.
- Experience in architecting complete scalable **data pipelines, data warehouse** for optimized data ingestion.
- Expertise in Built Enterprise ingestion **Spark framework** to ingest data from different sources **(s3, Salesforce, Excel, SFTP, FTP and JDBC Databases)** which is 100% metadata driven and 100% code reuse which lets Junior developers to concentrate on core business logic rather spark/Scala coding.
- Experience with **RESTful web services** and invoked them using **Postman.**
- Expertise in building microservice using **AWS lambda.**
- Good experience working with **Parquet files** and parsing, validating **JSON format files.**
- Experience on **NoSQL databases** like **MongoDB, Document DB** and **Graph Databases** like **neo4j.**
- Good Knowledge in working with **Flask framework** for designing **REST APIs** using Python language Flask Framework.
- Experience working on **pipelines** to engineer machine learning models using **Azure ML studio.**
- Expert in developing **ETL** jobs using Spark to SQL Database Systems and NoSQL database.
- Experience working on Continuous integration and continuous deployment using **Jenkins.**
- Hands-on experience on version control like **GitHub.**

TECHNICAL SKILLS:

Methodologies	Agile, SDLC Waterfall.
Languages	C, SQL, PL/SQL, Python, Shell Scripting, Scala, R, Core Java.

Bigdata Technologies	Hadoop MapReduce, Zookeeper, Hive, Pig, Sqoop, Yarn, Flume, Oozie, Storm, Spark, Kafka, HDFS, HBase, Impala, Zookeeper, NiFi, Spark Streaming, Machine Learning.
Web Technologies	jQuery, Node.js, JavaScript, HTML, CSS, Spring, J2EE, JDBC, Angular, Hibernate, Tomcat
Azure Cloud Services	Data factory, Azure Data Lake, Azure Databricks, Azure SQL database, Azure SQL Datawarehouse
AWS Cloud Services	AWS Lambda, AWS EC2, AWS API Gateway, EMR, S3, Glue, Data Pipeline, SNS, CloudWatch, Athena, AWS Databricks
Databases and Query Languages	Azure SQL Warehouse, Azure SQL DB, Azure Cosmos NoSQL DB, Teradata, Vertica, RDBMS, MySQL, Oracle, PostgreSQL, Microsoft SQL Server
Streaming Frameworks	Kinesis, Kafka, Flume.
Tools	PyCharm, Notebook, Eclipse, IntelliJ, NetBeans, Jupyter , R Studio, Teradata SQL Assistant, Autosys.
NoSQL	MongoDB, Cassandra, HBase.
ETL/BI	Power BI, Tableau, Talend, Snowflake, Informatica, SSIS, SSRS, SSAS
Reporting	Tableau.
Version control	GIT, SVN, Bitbucket.
Database	SQL Server, Document DB, MySQL, Neo4j, Teradata, Oracle, DB2. DynamoDB.
API Frameworks	Flask Framework (python).
Operating Systems	Windows, Linux, Unix.

Masters: **Aug 2021 – Dec 2022**

Masters in Computer Science, University of Dayton, OH | **CGPA :** 3.64

Professional Experience:

Client: Endeavour Technologies/MasterCard, MO, USA **April 2023 – Current**
Role: Data Engineer

Responsibilities:

- Wrote shell **script** to trigger data Stage jobs.
- Assisted service developers in finding relevant content in the existing reference models, like Access, Excel, CSV, Oracle, flat files using connectors, tasks and transformations provided by **AWS Data Pipeline.**
- Utilized **Spark** SQL API in **PySpark** to extract and load data and perform SQL queries.
- Worked on developing **Pyspark** script to encrypt the raw data by using hashing algorithms concepts on client specified columns.
- Designed, implemented, and maintained real-time data streaming pipelines using **Apache Kafka** for ingesting, processing, and analyzing high-volume data streams.
- Responsible for Design, Development, and testing of the database and Developed Stored Procedures, Views, and Triggers.
- I used **AWS Lambda** to execute the code for managing servers and containers for serverless computing service.
- Scaling for **AWS Lambda** is carried out immediately by gauging the workload attached to it.
- I integrated and used **AWS Lambda** in pipelines to react to web queries.
- Developed **Python**-based **API** (RESTful Web Service) to track revenue and perform revenue analysis.
- Designed and implemented end-to-end data pipelines using **Python** and Apache Spark, from data ingestion to transformation and loading into target databases, enabling real-time and batch data processing.
- Automated the provisioning and configuration of **AWS** resources with **Terraform**, reducing manual effort and ensuring consistency and repeatability in deployment processes.
- Worked on **AWS Data pipeline** to configure data loads from **S3** to **Redshift**.
- Used **AWS** Redshift, I Extracted, transformed, and loaded data from various heterogeneous data sources and destinations.
- Created Tables, Stored Procedures, and extracted data using **T-SQL** for business users whenever required.

- Deployed **Micro Services,** including provisioning **AWS environments** using Ansible Playbooks.
- Automated various infrastructure activities like **Continuous Deployment, Application Server setup, stack monitoring** using Ansible playbooks and has Integrated Ansible with **Jenkins.**
- Created dashboards for analyzing POS data using Power **BI.**
- Led the development of software solutions utilizing Agile and DevOps practices to implement data engineering workflows on the **Databricks** platform.
- **Used KPI** calculator Sheet and maintain that sheet within SharePoint.
- Created **Tableau** reports with complex calculations and worked on Ad-hoc reporting using **PowerBI**.
- Created a data model that correlates all the metrics and gives a valuable output.
- Worked on the tuning of **SQL** Queries to bring down run time by working on Indexes and Execution Plan.
- Performed **ETL** testing activities like running the Jobs, Extracting the data using necessary queries from database transform, and upload into the **Data warehouse** servers.
- Pre-processing using **Hive** and **Pig**.
- Extracted Transform and Load data from Sources Systems to **Azure Data Storage** services using a combination of **Azure** Data Factory, **T-SQL, Spark SQL,** and **U-SQL Azure Data Lake Analytics**.
- Worked in Data Ingestion to one or more Azure Services **-** (Azure Data Lake, Azure Storage, Azure SQL, and Azure DW) and processing the data in In **Azure Databricks.**
- Implemented Copy activity, Custom **Azure Data** Factory Pipeline Activities
- Primarily involved in Data Migration using **SQL, SQL Azure, Azure Storage, Azure Data Factory, SSIS, PowerShell.**
- Involved in Architect & implement medium to large scale **BI** solutions on Azure using **Azure Data** Platform services (Azure Data Lake, Data Factory, Data Lake Analytics, Stream Analytics, Azure SQL DW, HDInsight/Databricks, and NoSQL DB).
- Manage and administer Oracle, MySQL, SQL Server, and Apache **HBase** databases, ensuring high availability and performance.
- Optimized **Maven** configurations to improve build performance and reduce dependency conflicts.
- Created action filters, parameters and calculated sets for preparing dashboards and worksheets using **PowerBI.**
- Developed visualizations and dashboards using **PowerBI.**
- Used **ETL** to implement the Slowly Changing Transformation, to maintain Historically Data in Data warehouse.
- Performed **ETL** testing activities like running the Jobs, Extracting the data using necessary queries from database transform, and upload into the Data warehouse servers.
- Implemented **Spark Scala** and **PySpark using Data Frames, RDD, Datasets** and **Spark SQL** for processing data.
- Experience with **HBase** features such as coprocessors, filters, and compactions.
- Developed a detailed project plan and helped manage the data conversion migration from the legacy system to the target **snowflake** database.
- Deployed and managed Data Analytics and Engineering resources on **AWS** using **Terraform**, including Amazon S3 buckets, EC2 instances, AWS Lambda functions, and EMR clusters.
- Implemented ad-hoc analysis solutions using **Azure Data Lake Analytics/Store**, **HDInsight**
- Developed data pipeline using **Spark**, **Hive**, **Pig**, **python**, **Impala**, and **HBase** to ingest customers.
- Involved in converting **Hive/SQL** queries into **Spark** transformations using **Spark RDDs**, **Python** and **Scala**.
- Implemented **CI/CD** pipelines using **Jenkins** and GitLab CI/CD to automate testing, deployment, and monitoring of data engineering workflows, reducing manual effort and accelerating time to market.
- Worked on a direct query using **PowerBI** to compare legacy data with the current data and generated reports and stored and dashboards.
- Designed **SSIS** Packages to extract, transfer, load (**ETL**) existing data into **SQL** Server from different environments for the **SSAS** cubes (OLAP).
- Built performant, scalable **ETL** processes to load, cleanse and validate data.
- Implemented **Kafka** Streams for real-time data processing and stream analytics, enabling timely insights and decision-making.
- Used **Ansible Playbooks** to setup and configure **Continuous Delivery Pipeline** and **Tomcat servers.**

Environment: MS SQL Server, T-SQL, SQL Server Integration Services (SSIS), SQL Server Reporting Services (SSRS), SQL Server Analysis Services (SSAS), Management Studio (SSMS), Advance Excel (creating formulas, pivot tables, Hlookup, Vlookup, Macros), Spark, Python, ETL, Power BI, Tableau, Hive/Hadoop, Snowflakes, Power BI, AWS Data Pipeline, IBM Cognos, Data Stage, Cognos Report Studio, Cognos 8 & 10 BI, Cognos Connection, Cognos office Connection, Cognos, Data stage and Quality Stage.

Client: CSA Software Private Limited/Anthem, Indianapolis, Indiana, USA **Mar 2018 – July 2021**

Role: Data Engineer

Responsibilities:

- Used **Azure Data Factory** extensively for ingesting data from disparate source systems.
- Used Azure Data Factory as an orchestration tool for integrating data from upstream to downstream systems.
- Optimize **HBase** configurations and settings to improve query performance and resource utilization.
- Created numerous **pipelines** in **Azure** using **Azure Data Factory v2** to get the data from disparate source systems by using different Azure Activities like **Move Transform, Copy, filter, for each, Databricks etc.**
- Extract Transform and Load data from **Sources Systems** to **Azure Data Storage services** using a combination of **Azure Data Factory, T-SQL, Spark SQL** and **U-SQL Azure Data Lake Analytics.**
- Proficient in administering **HBase** clusters, including installation, configuration, and tuning.
- Data Ingestion to one or more Azure Services **- (Azure Data Lake, Azure Storage, Azure SQL, Azure DW)** and processing the data in **Azure Databricks.**
- Designed and developed user defined functions, stored procedures, triggers for Cosmos DB.
- Created interactive data visualizations and dashboards using **Python** libraries like Matplotlib, Seaborn, and Plotly to present insights and trends to stakeholders.
- Created DA specs and **Mapping Data flow** and provided the details to the developer along with **HLDs.**
- Knowledge of **Hadoop** ecosystem tools such as HDFS, MapReduce, and Apache ZooKeeper.
- Created Application Interface Document for the downstream to create a new interface to transfer and receive the files through **Azure Data Share.**
- Worked **Azure Databricks** to develop notebooks of PySpark and Scala for spark transformations
- Collaborated with cross-functional teams to design and implement event-driven architectures using **Kafka**, fostering a culture of real-time data-driven decision-making.
- Implemented **Maven** repositories to manage and distribute artifacts generated during the development process.
- Created several **Databricks Spark jobs** with **PySpark** to perform several tables to table operations.
- Extensively used **SQL Server Import** and **Export Data tool.**
- Ingested data in mini-batches and performs **RDD transformations** on those mini-batches of data by using **Spark Streaming** to perform streaming analytics in **Data bricks.**
- Developed **Kafka** Connectors to integrate various data sources and sinks, ensuring seamless data flow between systems.
- Implemented infrastructure as code best practices to maintain version control, modularity, and reusability of **Terraform** configurations, enabling efficient collaboration and scalability across projects.
- Performed ongoing monitoring, automation and refinement of data engineering solutions prepare complex **SQL views, stored procs** in **azure SQL DW** and **Hyperscale.**
- Designed and developed a new solution to process the **NRT data** by using **Azure stream analytics, Azure Event Hub** and **Service Bus Queue.**
- Design and implement **HBase** schemas based on application requirements and data access patterns.
- Worked with complex SQL, Stored Procedures, Triggers, and packages in large databases from various servers.
- Helping team members to resolve any technical issue, Troubleshooting, Project Risk & Issue identification and management.

Environment: Azure Cloud, Azure Data Factory (ADF v2), Azure functions Apps, Azure DataLake, BLOB Storage, SQL server, Teradata Utilities, Windows remote desktop, UNIX Shell Scripting, AZURE PowerShell, Data bricks, Python, Erwin Data Modelling Tool, Azure Cosmos DB, Azure Stream Analytics, Azure Event Hub, Azure Machine Learning.

Certifications: **Feb 2023 -Feb 2026**

AWS Certified Developer – Associate
VALIDATION NUMBER - LKW0HDP2R2VQQK5W
VALIDATE AT: https://aws.amazon.com/verification